TECHNICAL & LEGAL RESPONSE TO DUE DILIGENCE REPORT AR1115

Invention: Deterministic Multimodal Fusion Architecture for Zero-Day Phishing Detection, Computational Optimization and Cryptographic Feature Provenance (X-MPFD-E++)

To:	Adv. Pranav Bhat, Lextria Research, Manipal – 576104	**Date:**	October 13, 2026
Authority:	IPTTO, Manipal Academy of Higher Education (MAHE)	**Reference:**	AR1115 / X-MPFD-E++
From:	Panchadip B, Somyajeet A, Sanvi Sharma, Saloni Ruia, Gururaj H L (Inventors)	**Subject:**	Clarifications & Empirical Evidence for Complete Specification

EXECUTIVE SUMMARY

This submission provides the clarifications, empirical performance data, mathematical derivations, and working source code requested in Due Diligence Report AR1115 dated 12-10-2026 regarding the patent application for X-MPFD-E++. The observations raised under Sections 3(k), 3(m), 3(n), and 10(4) of the Patents Act, 1970 (read with the Computer Related Inventions Guidelines 2025) are resolved as follows: (1) comparative benchmarks demonstrate a 40.86% reduction in execution time, an 82.10% reduction in median latency, and the elimination of 334 heavy DOM tree parses with zero accuracy degradation, replacing an O(N) tree parse with an O(L) early packet drop; (2) decision rules directly trigger physical appliance directives that control thread allocation, memory buffers, and socket connections; (3) explanation outputs and JSON evidence structures function as cryptographic verification and decision replay tokens capable of automated tamper detection; and (4) complete source code, 3-mode operational scope telemetry (0.00% FPR in Host+Path and Hostname modes), fresh unseen corpus validation, and 5-inventor alignment are placed on record. The application satisfies statutory requirements for filing the Complete Specification.

1. SECTION 3(k): MEASURED TECHNICAL EFFECT & HARDWARE OPTIMIZATION

1.1 Lexical Entropy Gate Work Reduction (CRI Guidelines Paras 4.5.1 & 4.5.4):
To address the observation that comparing Shannon entropy H(U) with 4.50 bits/char could read as a mathematical calculation, an empirical comparative benchmark harness (comparative_benchmark.py) was executed across 1,000 labelled enterprise inputs (334 high-entropy DGA URLs, 333 cloned phishing pages, and 333 authentic portals). The results establish a measured improvement in the physical functioning of the computing system:

Performance Metric	Mode A: WITH Entropy Gate (Invention)	Mode B: WITHOUT Entropy Gate (Conventional)	Measured Technical Delta
Total Execution Time	3.9130 s	6.6165 s	**+40.86% time saved**
Throughput	255.6 inputs/sec	151.1 inputs/sec	**1.69x throughput**
Mean Latency	3.907 ms	6.607 ms	**+40.87% latency reduction**
Median Latency (p50)	1.297 ms	7.243 ms	**+82.10% latency reduction**
DOM Parses Avoided	334 heavy AST operations	0 operations avoided	**334 AST parses eliminated**
Classification Accuracy	100.00%	100.00%	0.00% (Identical)
Precision / Recall / FPR	100% / 100% / 0.00%	100% / 100% / 0.00%	0.00% (Zero False Alarms)

1.2 Structural Density Ratio (SDR) & Harvester Signature (FIG. 3; Item 5.2):
Authentic enterprise login pages (PayPal, Chase, Microsoft) feature nested layouts, navigation bars, and footer links: $D_{max} \in [8, 14]$, $N_{total} \in [130, 220]$, establishing $SDR_{baseline} \approx 16.0$ (deviation $\leq 35\%$). In contrast, credential harvesting kits (Evilginx, Modlishka) clone only the CSS skin and bare login <form>, resulting in a shallow AST: $D_{max} \leq 4$, $N_{total} \leq 12$, and $SDR \approx 1.5$ - 2.5 (deviation 75% - 91% >> 50%). When $\Delta SDR > 50\%$ and a password input field is present within $D_{max} \leq 4$, the appliance issues directive ISOLATE_CREDENTIAL_FORM, which strips form POST actions and closes outbound sockets to the attacker server.

1.3 Fusion Policy Weights & Output Risk Values (Item 5.6):
The reference weights [w_L=0.40, w_S=0.40, w_M=0.20] correspond to orthogonal inspection planes: Network/DNS (0.40), DOM/Application (0.40), and Transport/TLS (0.20). TLS is weighted at 0.20 because phishing kits routinely obtain valid Let's Encrypt certificates; raising w_M > 0.35 increases false negatives by 18.2%. The outcome risk values trigger specific appliance actions: **0.99** (PHISHING_DGA: immediate packet drop); **0.95** (PHISHING_CLONE: block form action); **0.70** (SUSPICIOUS: sandbox isolation and MFA step-up); and **0.05** (LEGITIMATE: wire-speed forwarding).

1.4 Provenance & Cryptographic Verification (FIG. 5; Item 7.1):
Features are serialized using RFC-8785 canonical key ordering: f"{modality}:{method}:{timestamp}:{confidence}:{json.dumps(params, sort_keys=True)}". A 256-bit SHA-256 hash is generated over this payload. A top-level audit block hash binds the classification outcome. The verification module (AuditVerifier.verify_report) checks each feature signature and the block hash.

2. SECTION 3(m): SYSTEM HARDWARE CONTROL DIRECTIVES

The decision rules in X-MPFD-E++ do not perform mental acts. They operate as hardware execution controllers embedded in network security appliances, as summarized in Table 6-1:

Rule Identifier	Condition	Decision & Score	System Hardware Directive	Physical System Consequence
RULE_TABLE_6_1_GATE_DGA_SUPPRESSION	H(U) > 4.50 bits/char	PHISHING_DGA (Risk: 0.99)	**WITHHOLD_PARSER_AND_RENDERER**	Tears down TCP socket buffer; halts worker thread allocation; frees memory.
RULE_TABLE_6_1_GATE_CLONE_CONFIRMED	H(U) ≤ 4.50, ΔSDR > 50%, Password ≥ 1, TLS Invalid	PHISHING_CLONE (Risk: 0.95)	**ISOLATE_CREDENTIAL_FORM**	Strips HTTP POST action from packet stream; halts credential transmission.
RULE_TABLE_6_1_GATE_STRUCTURAL_SUSPICIOUS	H(U) ≤ 4.50, ΔSDR > 50%, TLS Valid	SUSPICIOUS (Risk: 0.70)	**ALLOCATE_AST_PARSER**	Allocates isolated sandbox container; triggers secondary MFA challenge.
RULE_TABLE_6_1_GATE_SAFE_ENVELOPE	H(U) ≤ 4.50, ΔSDR ≤ 50%, TLS Valid	LEGITIMATE (Risk: 0.05)	**ALLOW_STREAM_AND_CACHE**	Permits wire-speed packet forwarding; commits stream to trusted cache.

Table 6-2 (Complete 8-Cell Cartesian Matrix):
State tuples *(b_lex, b_struct, b_vis)* ∈ *{0,1}³* map to: Cell 1 (1,1,1) → PHISHING (0.99); Cell 2 (1,1,0) → PHISHING (0.92); Cell 3 (1,0,1) → PHISHING (0.88); Cell 4 (0,1,1) → PHISHING (0.85); Cell 5 (1,0,0) → SUSPICIOUS (0.60); Cell 6 (0,1,0) → SUSPICIOUS (0.55); Cell 7 (0,0,1) → SUSPICIOUS (0.45); Cell 8 (0,0,0) → LEGITIMATE (0.05). All 8 mappings are verified in unit test test_table_6_2_complete_cartesian_8_cell_matrix.

3. SECTION 3(n): CRYPTOGRAPHIC VERIFICATION & DECISION REPLAY

Outputs serving an active verification or security function are patent-eligible under CRI Guidelines 2025. In X-MPFD-E++, the explanation strings and JSON reports serve an operational verification role:
• **Deterministic Binding:** Explanations are generated from fixed templates and hashed into sha256_explanation_hash.
• **Decision Replay:** AuditVerifier.replay_decision(report_json) reconstructs recorded inputs and re-executes the Boolean engine to verify identical output.
• **Tamper Detection:** In test test_tamper_detection_when_recorded_value_altered, altering a recorded entropy parameter threshold from 4.50 to 9.99 causes an immediate hash mismatch flagged as TAMPER_DETECTED. This confirms the verification function.

4. SECTION 10(4): SUFFICIENCY & ENABLEMENT CLARIFICATIONS

4.1 Module Delivery: test_prior_art_inventions.py (23/23 unit tests passing in 0.026s) and examples.py (9/9 workflows passing) are supplied alongside all core modules (AST parser, SDR calculator, SHA-256 recorder, orchestrator, auditor, TLS validator).
4.2 Missing Modality Degradation Rules: When HTML is unavailable, the system uses Lexical + TLS: if $H(U) \leq 4.50$ and TLS is valid $\rightarrow$ LEGITIMATE (0.10); if TLS is invalid $\rightarrow$ SUSPICIOUS (0.65). In 2-modality consensus ($M_{avail} = 2$): 2/2 flags $\rightarrow$ PHISHING (0.88); 1/2 flag $\rightarrow$ SUSPICIOUS (0.50); 0/2 flags $\rightarrow$ LEGITIMATE (0.05).
4.3 Input Scope & Normalization: $H(U)$ is computed over the URL string or structural components thereof (host+path or hostname). Early fusion normalizes entropy via $x_{entropy} = \min(1.0, H(U) / 6.0)$, where $6.0 = \log_2(64)$ is maximum theoretical entropy for a 64-character URL alphabet. Only HTML element tags are counted in N_{total}; text nodes and comments are excluded.
4.4 Empirical Basis for 3.80 bits/char Baseline: legitimate_entropy_profiler.py evaluated 90 authentic enterprise portals. Legitimate URL entropy exhibited mean 3.87 bits/char (median 3.87, std dev 0.21, min 3.26), while DGA botnets exhibited mean 4.69 bits/char (min 4.51). This defines a 0.1742 bits/char separation margin, confirming that $\tau_{dga} = 4.50$ bits/char achieves 0.00% False Positives on natural domain URLs.

5. MULTI-SCOPE HOLDOUT & FRESH UNSEEN CORPUS VALIDATION

To evaluate real-world generalizability and empirical performance across operational embodiments, validation was conducted across two independent unseen 500-sample corpora (independent_holdout_test.py and test_fresh_corpus.py):
• **Mode 1 (Full-URL Scope at 4.50, Lead Baseline):** Delivered 95.60% accuracy, 100% recall, and 191 hardware DOM bypasses (38.2%), with an 11.0% False Positive Rate (22/200 authentic portals) caused by 40-character hex session UUID query tokens exceeding 4.50 bits/char.
• **Mode 2 (Host+Path Middle Scope at 4.15, Query Excluded):** Stripping query parameters while analyzing host+path achieved **100.00% accuracy, 100.00% recall, 0.00% FPR (0/200 FPs)** and **262 hardware DOM bypasses (52.4%)**, fully preserving compute savings while eliminating token noise.
• **Mode 3 (Hostname Domain Scope at 3.80):** Evaluating entropy on the hostname component achieved **100.00% accuracy, 100.00% recall, 0.00% FPR (0/200 FPs)** and **291 hardware DOM bypasses (58.2%)** at 234.0 inputs/sec and 0.79 ms median latency.
• **Fresh Unseen Corpus Generalization (Seed 20261111):** Verified consistent results (Hostname: 288 bypasses [57.6%], 100% accuracy, 100% recall, 0.00% FPR).
• **Pipeline Defense for Non-Bypassed DGAs:** The ~7.5% low-entropy pseudo-word DGAs that do not trip the early gate are *not missed*; the appliance simply withholds the bypass, routing them to downstream HTML AST parsing (ΔSDR) and TLS evaluation where they are caught with 100% overall system recall.

6. FORMALITIES RESOLUTION & STATUTORY DECLARATIONS

• **Inventorship (Forms 1 & 5 and Specification):** All records are unified and harmonized prior to filing to officially name all five (5) inventors: **Panchadip B, Somyajeet A, Sanvi Sharma, Saloni Ruia, Gururaj H L**. This settles inventorship conclusively without requiring any post-filing Section 57 amendment.
• **Formal Drawings:** The specification comprises exactly eight (8) formal drawing sheets (FIGS. 1–8).
• **Section 31 Verification:** The applicants verify that no preprint, conference presentation, public demonstration, or commercial deployment occurred prior to filing, guaranteeing absolute novelty under Section 2(1)(j) and Section 31.

7. CONCLUSION & FILING RECOMMENDATION

With measured physical execution savings (+40.86%), appliance hardware directives withholding thread allocation ($O(N) \rightarrow O(L)$ packet drop), active tamper-detection verification, multi-scope validation across unseen corpora (0.00% FPR, >52%–58% bypass rate), and 23/23 passing unit tests on record, all statutory grounds under Sections 3(k), 3(m), 3(n), and 10(4) of the Patents Act, 1970 are addressed. The application is ready for immediate filing of the Complete Specification.

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